

AHT Root Cause Analysis

Identifying Factors Associated with High Average Handle Time

Tools: Python | Pandas | Tableau | Excel

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Business Problem

Why is Average Handle Time not reducing?

- High AHT can reduce agent productivity.
- Longer interactions require more operational effort.
- Certain chat types may require more time to resolve.
- Identifying the major AHT drivers can help prioritize improvement opportunities.

Project Objective

Identify the key factors associated with high AHT and recommend actionable improvement opportunities.

Areas analyzed

Customer

- Customer response time
- Repeat messages
- Chat category

Agent

- Agent response time
- Navigation time
- Applications used

Operational

- Escalations
- Technical errors
- Shift

Dataset & Technology

Dataset

Synthetic customer-service dataset created for analytics practice.

10,000 customer chat records

Key fields:

- Chat Category
- AHT
- Agent Response Time
- Customer Response Time
- Navigation Time
- Repeat Messages
- Applications Used
- Escalation
- Technical Error
- CSAT
- Shift

Tools

Excel

-Dataset

Python

-Data cleaning & analysis

Tableau

-Interactive Dashboard

Analytical Approach

Data Understanding



Exploratory Analysis



Root Cause Analysis



Identify AHT Drivers



Tableau Dashboard



Recommendations

Module 1

Understand the data

Module 2

Explore AHT patterns

Module 3

Investigate potential drivers

Module 4

Visualize & communicate findings

Overall Performance

KPI	Result
Average AHT	~1,027.5 sec
Average CSAT	2.61 / 5
Total Chats	10,000
Escalation Rate	9.9%

Which Chat Categories Have the Highest AHT?

Category	AHT
Fraud	1,148 sec
Transaction Dispute	1,125 sec
Payment Dispute	1,109 sec
Hardship	1,094 sec
Promise to Pay	989 sec
Payment	981 sec
Debit Card	979 sec
Checking Account	970 sec
Wallet	958 sec
Disconnected	920 sec

Key AHT Drivers

Repeat Messages

1 → 930 sec

4 → 1,155 sec

+225 sec

Navigation Time

0–25 sec → 926 sec

40+ sec → 1,170 sec

+244 sec

Agent Response Time

20–30 sec → 899 sec

51–60 sec → 1,150 sec

+251 sec

Customer Response Time

40–50 sec → 913 sec

71–80 sec → 1,146 sec

+233 sec

Operational Factors

Escalation

No

1,014 sec

Yes

1,150 sec

Difference: ~136 sec

Technical Error

No

1,014 sec

Yes

1,121 sec

Difference: ~107 sec

Conclusion:

Escalation and technical errors are associated with higher AHT, while shift differences appear relatively small.

Key Insights

1. Fraud chats have the highest average AHT.
2. AHT increases as repeat messages increase.
3. Higher navigation time is strongly associated with higher AHT.
4. Higher agent and customer response times are associated with higher AHT.
5. Escalated chats have higher AHT than non-escalated chats.

Overall: AHT appears to be more strongly associated with interaction and process-efficiency factors than with shift.

Recommendations

1. Reduce Navigation Time

Review workflows requiring multiple applications and identify opportunities to simplify system navigation.

2. Improve Response Efficiency

Create standardized guidance and knowledge resources for complex interactions.

3. Reduce Repeat Messages

Identify common customer confusion points and improve first-response clarity.

4. Review Escalation Drivers

Identify frequently escalated scenarios and provide agents with better resolution guidance.

5. Prioritize High-AHT Categories

Fraud → Transaction Dispute → Payment Dispute → Hardship

Interactive Tableau Dashboard

The dashboard enables users to:

- Monitor AHT KPIs
- Compare AHT by chat category
- Analyze key AHT drivers
- Filter results interactively
- Identify areas for operational improvement

AHT Root Cause Analysis Dashboard

Objective: Identify the key drivers of high Average Handle Time (AHT) and recommend improvement opportunities using Python and Tableau.

Average AHT (sec)

1,027.5

Average CSAT/5

2.611

Total Chats

10,000

Escalation Rate

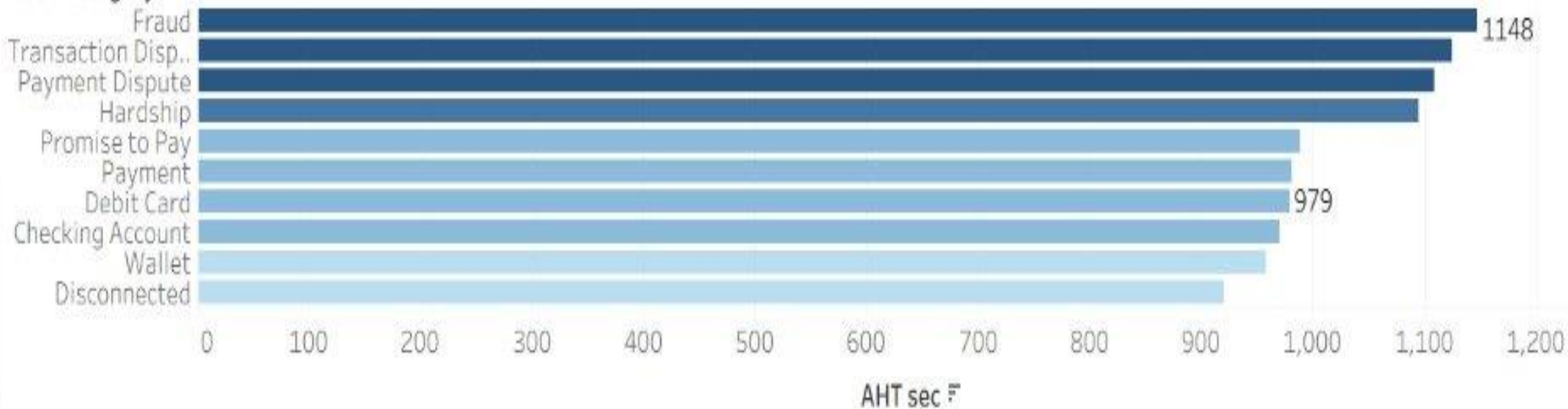
9.9%

Chat Category

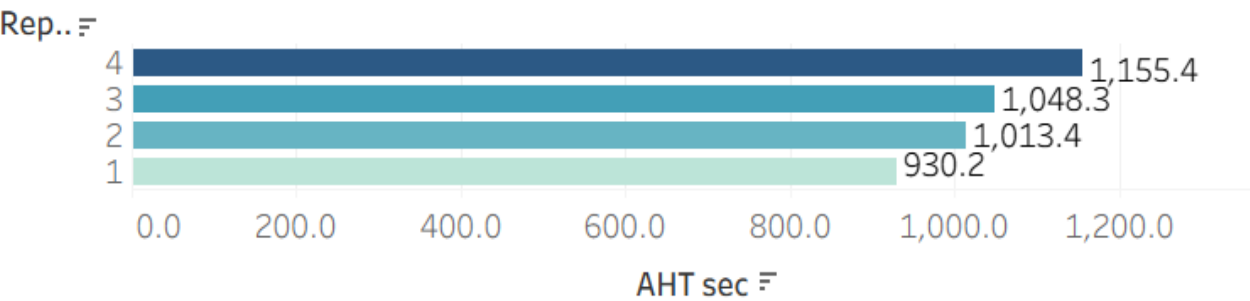
- ☒ (All)
- ☒ Checking A...
- ☒ Debit Card
- ☒ Disconnect...
- ☒ Fraud
- ☒ Hardship
- ☒ Payment
- ☒ Payment D...
- ☒ Promise to...
- ☒ Transactio...
- ☒ Wallet

Average AHT by Chat Category

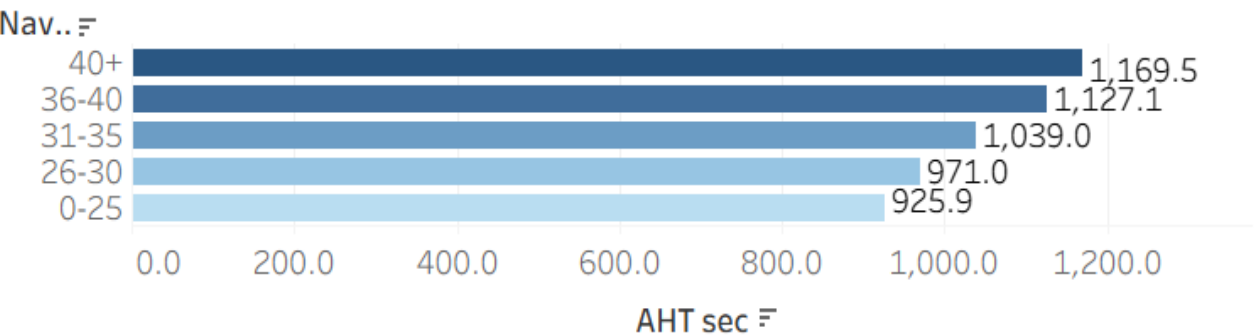
Chat Category =



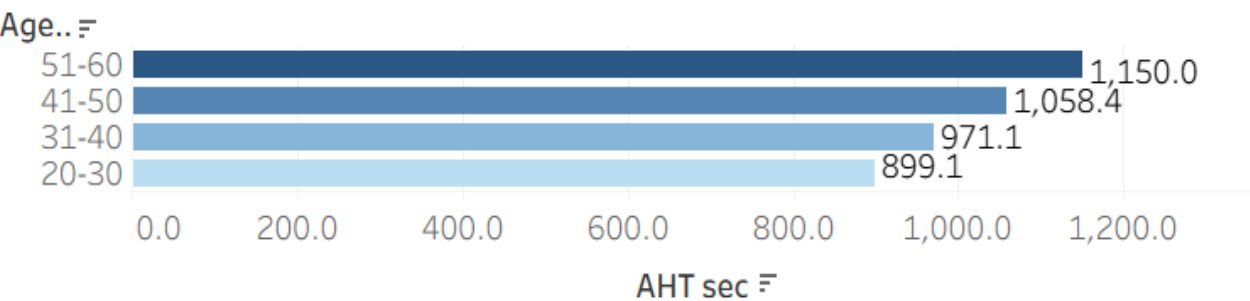
AHT Increases with Repeated Messages



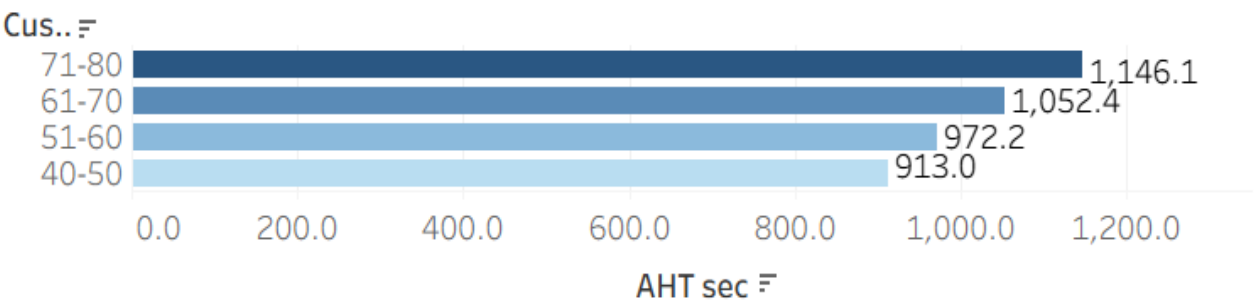
AHT Increases with Navigation Time



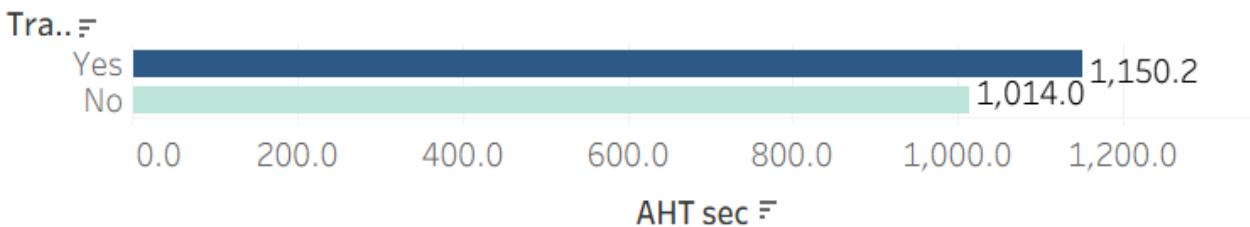
AHT Increases with Agent Response Time



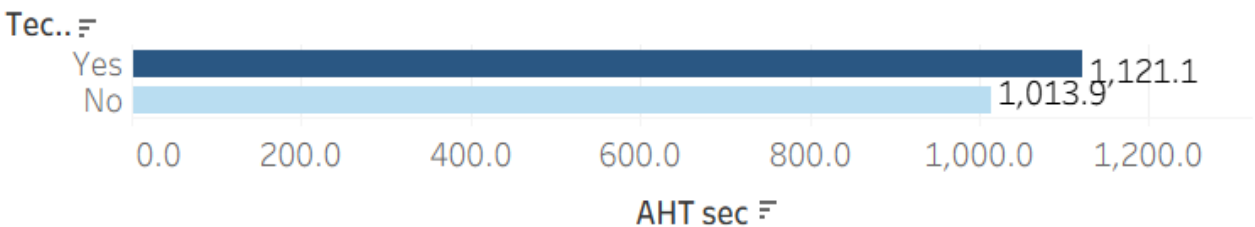
AHT Increases with Customer Response Time



Escalated Chats Have Higher AHT



Technical Errors Are Associated with Higher AHT



Conclusion

AHT is not driven by a single factor.

Higher AHT is consistently associated with:

- Navigation time
- Agent response time
- Customer response time
- Repeat messages
- Escalation

Shift differences appear relatively small.

The analysis provides a data-driven starting point for improving customer service efficiency and reducing unnecessary handling time.